

Broad Medical Knowledge Base

Educational reference document for the MEDEASY retrieval-augmented generation system.

1. General Medicine and Clinical Reasoning

Clinical assessment commonly combines history, physical examination, laboratory testing, and imaging. Important clinical concepts include symptoms, signs, differential diagnosis, risk factors, disease course, treatment response, and follow-up. A single symptom or laboratory value usually cannot establish a diagnosis by itself. Clinical interpretation should consider the patient's age, history, medications, examination findings, and relevant reference ranges.

2. Cardiology

Cardiovascular diseases include hypertension, coronary artery disease, heart failure, arrhythmias, valvular disease, and cardiomyopathies. Common cardiovascular risk factors include smoking, diabetes, dyslipidemia, obesity, physical inactivity, and family history. Evaluation may include blood pressure measurement, electrocardiography, echocardiography, cardiac biomarkers, stress testing, and vascular imaging. Heart failure may produce symptoms such as breathlessness, fatigue, and peripheral edema. Chest pain, severe breathlessness, fainting, or other acute symptoms require prompt medical assessment.

3. Neurology

Neurological disorders include stroke, epilepsy, headache disorders, Parkinson disease, dementia, multiple sclerosis, neuropathies, and brain tumors. Neurological assessment considers mental status, cranial nerves, strength, sensation, coordination, reflexes, gait, and speech. MRI and CT are important imaging methods for evaluating neurological conditions. Stroke is a medical emergency in which rapid recognition and treatment are important.

4. Respiratory Medicine

Respiratory conditions include asthma, chronic obstructive pulmonary disease (COPD), pneumonia, tuberculosis, pulmonary embolism, interstitial lung disease, and respiratory infections. Symptoms may include cough, wheezing, sputum production, chest discomfort, and shortness of breath. Evaluation can include pulse oximetry, chest radiography, CT, spirometry, blood tests, and microbiological testing. Severe breathing difficulty or low oxygen saturation requires urgent evaluation.

5. Endocrinology and Diabetes

The endocrine system includes glands and hormones that regulate metabolism, growth, reproduction, stress responses, and other processes. Diabetes mellitus is characterized by abnormal blood glucose regulation and includes type 1 and type 2 diabetes among other forms. Common assessments include fasting glucose, random glucose, and glycated hemoglobin (HbA1c). Thyroid disorders include hypothyroidism, hyperthyroidism, thyroid nodules, and thyroiditis. Thyroid evaluation commonly uses thyroid-stimulating hormone (TSH) and free thyroid hormone measurements together with clinical assessment.

6. Gastroenterology and Liver Disease

Gastrointestinal disorders include gastroesophageal reflux disease, peptic ulcer disease, inflammatory bowel disease, irritable bowel syndrome, gastrointestinal infections, and colorectal disease. Liver disorders include fatty liver disease, viral hepatitis, cirrhosis, and liver tumors. Laboratory evaluation may include AST, ALT, alkaline phosphatase, bilirubin, albumin, and coagulation studies. Ultrasound, CT, MRI, and endoscopy can provide additional diagnostic information depending on the clinical question.

7. Nephrology

Kidney disorders include acute kidney injury, chronic kidney disease, nephritis, kidney stones, and urinary tract disorders. Kidney function is commonly assessed using serum creatinine and estimated glomerular filtration rate (eGFR), while urine testing can identify protein, blood, glucose, infection, and other abnormalities. Chronic kidney disease may progress over time and is associated with conditions such as diabetes and hypertension.

8. Oncology

Cancer is a group of diseases characterized by uncontrolled cell growth and the potential to invade surrounding tissues or spread to distant organs. Major cancer types include breast, lung, colorectal, prostate, skin, liver, and hematologic cancers. Diagnosis may involve imaging, laboratory tests, endoscopy, and pathology. Treatment can include surgery, radiation therapy, chemotherapy, targeted therapy, immunotherapy, hormonal therapy, or combinations of these approaches. Treatment depends on tumor type, stage, molecular characteristics, and patient factors.

9. Infectious Diseases

Infectious diseases are caused by pathogens including bacteria, viruses, fungi, and parasites. Diagnosis may involve clinical assessment, cultures, antigen tests, molecular tests such as PCR, serology, and imaging. Antimicrobial treatment depends on the causative organism and susceptibility when known. Antibiotics do not treat viral infections. Infection prevention can include vaccination, hand hygiene, respiratory precautions, food and water safety, and appropriate isolation measures.

10. Dermatology

Common dermatological conditions include acne, eczema, psoriasis, dermatitis, fungal infections, bacterial infections, viral skin infections, and skin cancers. Assessment considers lesion morphology, distribution, duration, symptoms, exposures, and associated systemic findings. Some lesions require dermoscopy or biopsy for definitive diagnosis. Changes in a pigmented lesion such as asymmetry, irregular borders, varied color, increasing size, or evolving appearance warrant professional evaluation.

11. Hematology

Hematology covers disorders of blood cells, bone marrow, and coagulation. Anemia can result from blood loss, nutritional deficiency, chronic disease, hemolysis, or impaired blood-cell production. A complete blood count (CBC) evaluates hemoglobin, hematocrit, red-cell indices, white blood cells, and platelets. Iron studies, vitamin levels, reticulocyte count, peripheral smear, and other tests may help determine the cause of anemia.

12. Musculoskeletal Medicine

Musculoskeletal conditions include osteoarthritis, rheumatoid arthritis, osteoporosis, fractures, tendon disorders, ligament injuries, and back pain. Evaluation can include physical examination and imaging such as X-ray, ultrasound, CT, or MRI. Bone health assessment may include bone mineral density testing. Treatment varies by condition and can include physical therapy, exercise, medicines, activity modification, and surgery when indicated.

13. Psychiatry and Mental Health

Mental health conditions include depressive disorders, anxiety disorders, bipolar disorder, schizophrenia-spectrum disorders, substance-use disorders, and trauma-related disorders. Assessment commonly includes symptoms, duration, functional impact, medical history, medications, substance use, and psychosocial factors. Treatment can involve psychotherapy, lifestyle interventions, medications, social support, or combinations. Mental-health emergencies involving immediate danger require urgent professional assistance.

14. Medical Laboratory Tests

Laboratory tests provide measurements that support clinical assessment but should be interpreted in context. Common tests include CBC, electrolytes, glucose, HbA1c, renal function, liver enzymes, lipid profile, thyroid tests, inflammatory markers, urinalysis, and microbiological tests. Reference ranges vary between laboratories and patient populations. Abnormal results do not automatically indicate a disease and may require repeat testing or additional evaluation.

15. Medical Imaging

X-ray uses ionizing radiation to create projection images and is widely used for bones and chest imaging. CT produces cross-sectional images using X-rays and can provide detailed anatomical information. MRI uses magnetic fields and radiofrequency energy and is particularly useful for soft tissues, the brain, spine, joints, and many abdominal structures. Ultrasound uses sound waves and is useful for many abdominal, vascular, obstetric, cardiac, and superficial structures. Imaging findings should be interpreted together with clinical information.

16. Brain MRI and Brain Tumors

Brain tumors are abnormal growths arising within or around the brain. Primary tumors include gliomas, meningiomas, pituitary tumors, and schwannomas, while metastases originate from cancers elsewhere. MRI is commonly used to characterize intracranial lesions. Symptoms depend on location and may include headache, seizures, visual changes, weakness, balance problems, or cognitive changes. Treatment depends on tumor type, grade, location, molecular characteristics, and patient factors and may involve surgery, radiation, systemic therapy, or supportive care.

17. Deep Learning in Medical Imaging

Deep learning methods such as convolutional neural networks (CNNs), transfer learning, and U-Net-style segmentation networks are widely investigated in medical image analysis. Applications include classification, detection, segmentation, and image enhancement. Evaluation can use sensitivity, specificity, precision, recall, F1 score, accuracy, and ROC-AUC. Important limitations include dataset bias, class imbalance, scanner variation, image quality, labeling errors, data leakage, and differences between research datasets and clinical populations.

18. Chest X-ray AI and Respiratory Imaging

Artificial intelligence systems have been investigated for chest X-ray classification and detection of findings such as pneumonia and other pulmonary abnormalities. CNNs can learn visual patterns, while transfer learning can help when labeled datasets are limited. Research evaluation commonly includes sensitivity, specificity, precision, recall, F1 score, and ROC-AUC. AI outputs should be considered decision-support information unless a system has undergone appropriate validation and authorization for clinical use.

19. Preventive Health

Preventive medicine includes vaccination, screening, healthy nutrition, physical activity, tobacco avoidance, appropriate alcohol use, sleep, and management of cardiovascular and metabolic risk factors. Screening recommendations depend on age, sex, risk factors, family history, and local clinical guidelines. Prevention strategies should be individualized with qualified healthcare professionals when necessary.

20. Safety and Interpretation

This document is an educational knowledge-base resource for information retrieval. It is not a diagnostic guideline and should not be used as a substitute for professional medical evaluation. Medical recommendations can change as evidence and clinical guidelines are updated. When information is incomplete, uncertain, or potentially urgent, professional medical assessment should be sought.